

Android Studio (ako začat')

Jazyk Kotlin (ako neskončit')



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KAI, I-18

MS-Teams: [2sf3ph4](#), [List](#), [github](#)

borovan 'at' ii.fmph.uniba.sk



Základné info o kurze

■ Stránka predmetu

- <https://dai.fmph.uniba.sk/courses/VMA/>

prihláste sa do L.I.S.T.

- ak ste v ňom nikdy neboli, ozvite sa mi mailom
- sledujte LIST, všetky zadania budú v ňom
- sledujte Teams [2sf3ph4](#), komunikácia/prednášky/oznamy



veľká časť kurzu bude dobre sledovateľná z knihy

- **Android Studio Quail Essentials - Kotlin Edition: Developing Android Apps Using Android Studio and Kotlin , Neil Smyth** <https://www.payloadbooks.com/product/android-studio-koala-essentials-kotlin-edition-ebook/>
- .pdf pre staršiu verziu Giraffe je k dispozícii... <https://www.amazon.com/Android-Studio-Giraffe-Essentials-Developing/dp/1951442776>
- **Android Studio Quail Essentials - Compose Edition: Developing Android Apps with Android Studio, Jetpack Compose, and Kotlin,** https://www.amazon.com/Android-Studio-Quail-Essentials-Developing/dp/1965764436/ref=sr_1_1?dib=eyJ2IjojMSJ9.RV7TfYX4py0FmL-jSTnU01d-cZjhA7RPIi1sA0M8J9jn7gIm-bdiMcWGRZbQC3Sk-ITVgmSHKFw7VJUx2vksCeeijv065Y10sMYiqyzfyP92uQc0V4D-n4K5xNHJxxAoDTtdMwh0INcaVmejCo214QqRyFEeMOLCU7KYnxUyxe83IF2cJSauxmKftnSOOGZFX3wJJMxWAAAZc-RH4L5hhVynytfOhK9TbwSS0sZB0Y.RZer1DK1jxJ04mQauGGiC0ifGruk9ms7zJv4aHRIpEE&dib_tag=se&keywords=Android+Studio+Development+Essentials+-+Neil+Smyth&qid=1786614317&s=books&sr=1-1

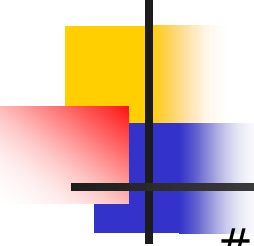
Vývojové jazyky/nástroje

- Symbian
 - C++, Java ME, Python, ...
- Windows Mobile 6
 - C# (MS Visual Studio)
- iOS
 - Objective-C -> Swift 3/4/5 (Xcode)
- Android
 - scratch (MIT Inventor)
 - java (Android SDK + plugin pre Eclipse) -> (Android Studio)
 - java (A.I.D.E.)
 - Kotlin (Android Studio 4+)
 - C++ (Android NDK)
- Multi-platform
 - C# (Xamarin iOS, Android, Windows) – fy. MS, Visual Studio 2015
 - Pascal (Delphi XE5 iOS, Android, Windows 10) – fy. Embarcadero
 - JavaScript/TypeScript (React Native)
 - Flutter od Google
- game engine
 - C# (Unity 2D/3D), C++ (Unreal Engine)

iOS - Apple Center kurz

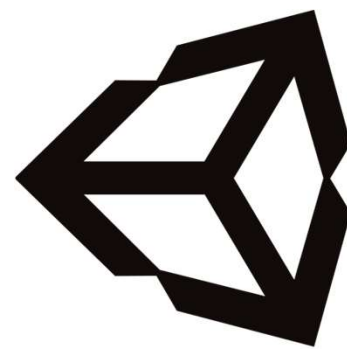
1-AIN-303/24

...



Unity

(game engine)



unity

#1 Unreal Engine

#2 Unity

Oblíbený nástroj pre tvorbu multi-platform aplikácií pre bakalárske práce

- 2-INF-263/15 magisterský predmet: Tvorba a dizajn počítačových hier
 - <http://sccg.sk/~mferko/tdh/>
 - <https://candle.fmph.uniba.sk/ucitelia/Michal-Ferko>
- 1-AIN-303/24 bakalársky predmet: Game Engines
 - Šimko

Vývoj a nástroje

(detailnejšie)

■ natívne aplikácie

- Android



- Java
- Kotlin



- iOS



- Objective-C
- Swift



Priamy prístup k všetkým fičúrkam a komponentom OS, aj tým najnovším ...

■ hybridné aplikácie

- Cordova

- ionic



APACHE
CORDOVA™



Web-app na báze .html, .css, .js, ktoré púšťame v prostredí WebView=browser/wrapper bez browserových ovládačov

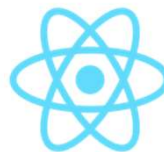
Vývoj a nástroje

(detailnejšie)

■ kompilované aplikácie

■ React Native

- JavaScript
- Facebook
- nekompiluje do natívneho kódu
- obmedzená množina widgets



■ Flutter (Dart framework od Googlu)

- Dart
- Google
- kompilované do ARM C++
- bohatšia množina widgets
- Material design (Quantum Paper)– Google 2014



■ NativeScript

- JavaScript



QUANTUM
and the building blocks of a unified interface



Aspekty programátora

Code sharing (write once, use everywhere)

- Cordova, ionic
- Flutter (Material Design)
- ReactNative
- Java, Kotlin, Swift



Knowledge sharing (learn once, use everywhere)

- Cordova, ionic, ReactNative (.js), Flutter (Dart)
- Java, Kotlin, Swift



Widget library

- Java, Swift, Cordova, ionic
- Flutter (Dart)
- ReactNative (.js)



<https://www.youtube.com/watch?v=bnYJRYFsrSw&feature=youtu.be>

Aspekty programátora

Eco-system (schopnosť nájsť riešenie/radu/blog na stackoverflow,..)

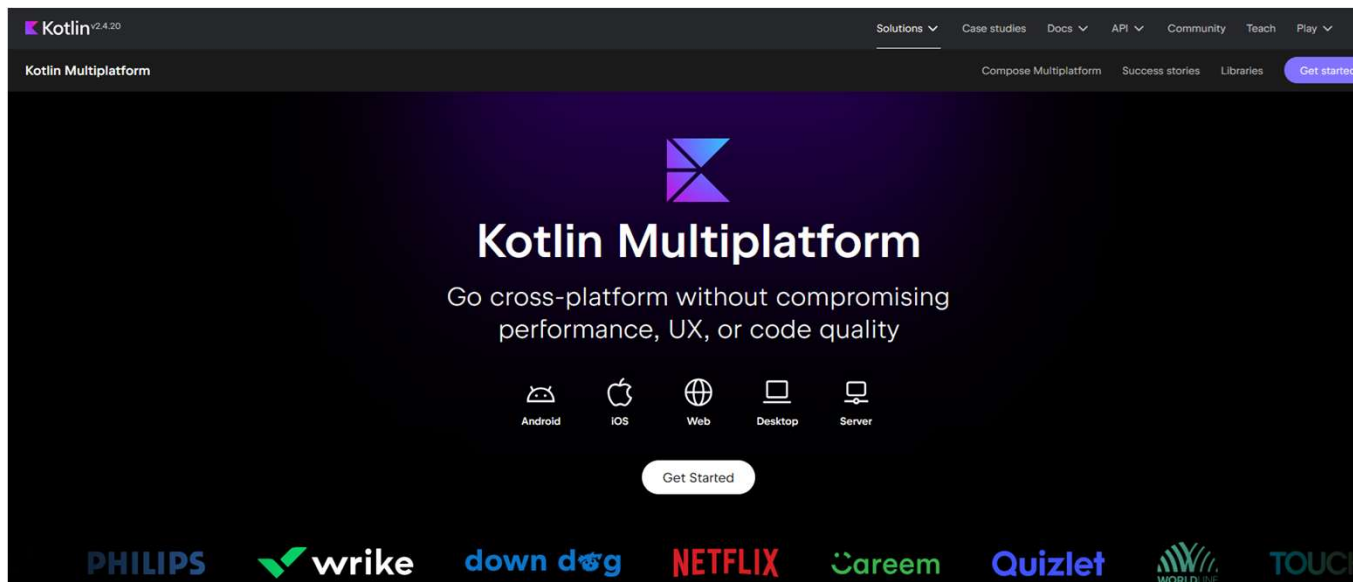
- Java, Swift, Kotlin 😊
- Cordova, ionic (.js) 😐
- ReactNative (.js, React) 😐
- Flutter (nové ale zlepšuje sa) 😞

Popularita

- Java, Swift, Kotlin 😊
- Cordova, ionic, ReactNative, Flutter (pushujú FB a Google) 😐

Kotlin Multiplatform

- *KMM* is an SDK designed to simplify creating cross-platform mobile applications (Android, iOS iPhone, watchOS, Windows, Linux)
- share common code between iOS and Android apps
- write platform-specific code
- platforms jvm, js, wasm
- xcode (mac) is necessary to build an iOS app



Java vs. Kotlin



tradičný VMA kurz postavený na Java už šiesty rok beží v jazyku Kotlin

Dôvody:

- ako iOS má svoj moderný jazyk Swift (3/4/5), aj Android má svoj Kotlin
 - Java je trochu *skamenelina* medzi modernými jazykmi (Swift, Kotlin, Scala)
 - Kotlin je Googlom oficiálne podporovaným vývojový nástroj pre Android
 - projekt Kotlin má už >15 rokov
 - kompiluje do JVM
 - funguje s Android Studiom
 - na JetBrains produkty ste si asi zvykli, a sú top
 - oboznámite sa s niektorými princípmi moderných jazykov
-
- Reference: <https://kotlinlang.org/docs/reference/>
 - Online: <https://play.kotlinlang.org/byExample/>

Android Studio Quail Kotlin Edition



- <https://www.payloadbooks.com/product/android-studio-koala-essentials-kotlin-edition-ebook/>
- <https://www.amazon.com/Android-Studio-Giraffe-Essentials-Developing/dp/1951442776>
- <https://www.ebookfrenzy.com/errata/giraffekotlin.html>
- sources: <https://www.ebookfrenzy.com/retail/giraffekotlin/index.php>

Inštalácia Android Studio:

<https://developer.android.com/studio>

Android Studio

The official IDE for Android app development now accelerates your productivity with Gemini in Android Studio, your AI-powered coding companion.

Download Android Studio Quail 3 

Predmet má cvičenie, ale aj tak:

- ozvite sa v prípade problémov inštalácie na platformy, napr. Linux, Mac.
- posnažíme problém vyriešiť

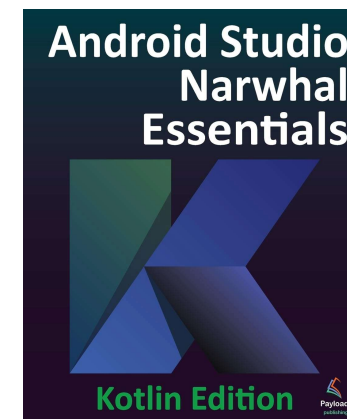
Inštalácia Android Studia:

- SDK Packages: Tools/SDK Manager tab SDK Platforms
- AVD: Android Virtual Device

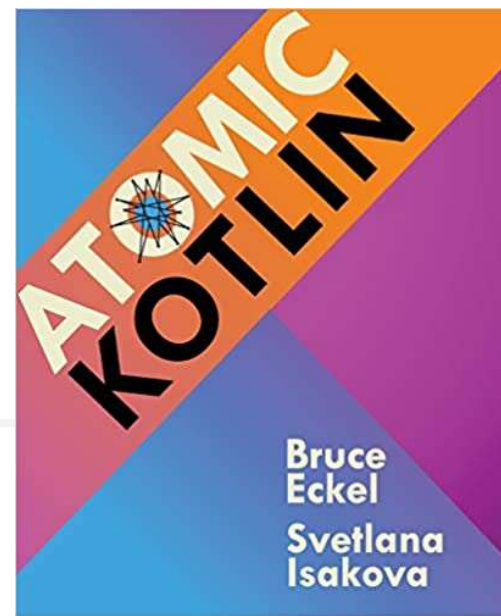
Android Studio Quail Kotlin Edition



- <https://www.payloadbooks.com/product/android-studio-koala-essentials-kotlin-edition-ebook/>
 - <https://www.amazon.com/Android-Studio-Giraffe-Essentials-Developing/dp/1951442776>
 - <https://www.ebookfrenzy.com/errata/giraffekotlin.html>
 - sources: <https://www.ebookfrenzy.com/retail/giraffekotlin/index.php>
2. Setting up an Android Studio Development Environment (mac/Windows/Linux)
 3. Creating an Example Android App in AS
 4. Creating an Android Virtual Device (AVD) in AS
 5. Using and Configuring the Android Studio AVD Emulator
 6. A tour of the Android Studio User Interface
 7. Testing Android Studio App on a Physical Android Device
 8. The Basics of the Android Studio Code Editor.
 9. An Overview of the Android Architecture
 10. The Anatomy of an Android App
 11. An Introduction to Kotlin
 12. Kotlin Data Types, Variables, and Nullability
 13. Kotlin Operators and Expressions
 14. Kotlin Control Flow
 15. An Overview of Kotlin Functions and Lambdas
 16. The Basics of Object Oriented Programming in Kotlin
 17. An Introduction to Kotlin Inheritance and Subclassing
 - ...
 91. An Overview of Gradle in Android Studio

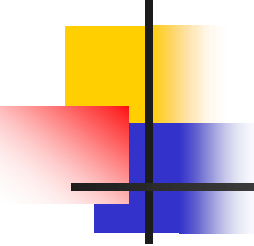


Atomic Kotlin



<https://www.amazon.com/Atomic-Kotlin-Bruce-Eckel/dp/0981872557>

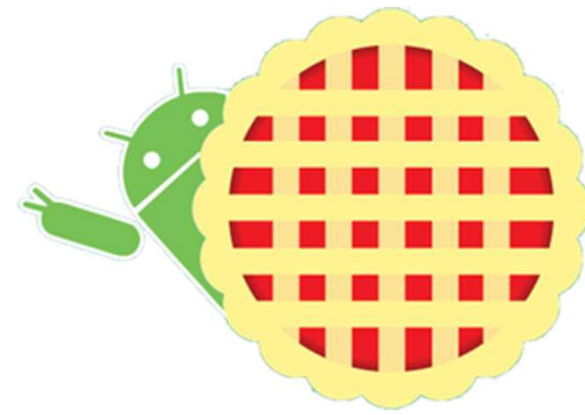
Section I: Programming Basics			
Introduction			
Why Kotlin?			
Hello, World!			
var & val			
Data Types			
Functions			
if Expressions			
String Templates			
Number Types			
Booleans			
Repetition with while			
Looping & Ranges			
The in Keyword			
Expressions & Statements			
Summary 1			
	Section II: Introduction to Objects		
	Objects Everywhere		
	Creating Classes		
	Properties		
	Constructors		
	Constraining Visibility		
	Packages		
	Testing		
	Exceptions		
	Lists		
	Variable Argument Lists		
	Sets		
	Maps		
	Property Accessors		
	Summary 2		
		Section III: Usability	
		Extension Functions	
		Named & Default Arguments	
		Overloading	
		when Expressions	
		Enumerations	
		Data Classes	
		Destructuring Declarations	
		Nullable Types	
		Safe Calls & the Elvis Operator	
		Non-Null Assertions	
		Extensions for Nullable Types	
		Introduction to Generics	
		Extension Properties	
		break & continue	
			Section IV: Functional Programming
			Lambdas
			The Importance of Lambdas
			Operations on Collections
			Member References
			Higher-Order Functions
			Manipulating Lists
			Building Maps
			Sequences
			Local Functions
			Folding Lists
			Recursion
			Section V: Object-Oriented Programming
			Interfaces
			Complex Constructors
			Secondary Constructors
			Inheritance
			Base Class Initialization
			Abstract Classes
			Upcasting
			Polymorphism
			Composition
			Inheritance & Extensions
			Class Delegation
			Downcasting
			Sealed Classes



Why Teach Kotlin

- Kotlin is basically becoming the language of Android.
- Decrease in boilerplate helps us to quickly identify which fundamental Android concepts students are missing.
- On numerous courses, where we proceed through Java to Kotlin, we are considering a Kotlin-first approach.
- Students are happy to have the chance to program in something they may have heard about.
- I think our students benefit in general from being exposed to a wide range of programming languages, and I think it is valuable for them to gain experience in using more modern languages alongside the more traditional ones like Java and C++.
- My Kotlin students in fact understand OO concepts better than my Java students do.
- One of Kotlin's advantages is a good combination of strong typing and nullability.

Android a Google



2005 [Google](#) acquired Android Inc. with Rubin, Miner et al.

- 2007 [Open Handset Alliance](#), a consortium
 - device manufacturers: [HTC](#), [Sony](#) and [Samsung](#),
 - wireless carriers: [T-Mobile](#), ...
 - chipset makers: [Qualcomm](#), [Texas Instruments](#),
 includes Google with a goal to develop open standards for mobile devices
- major release named in alphabetical order after a dessert or sugary treat
 - 2.3 [Gingerbread](#)
 - 4.3 *Jelly Bean*, July, 2012,
 - 4.4 [KitKat](#), announced, October, 2013,
 - 5.1 [Lollipop](#), November, 2014,
 - 6.0 [Marshmallow](#), October, 2015,
 - 7.0 [Nougat](#), August, 2016.
 - 8.0 [Oreo](#), August, 2017,
 - 9.0 [Pie](#), August, 2018,
 - 10.0 [Android 10](#), September 2019
 - 11.0 [Android 11](#), ...

Android Oreo	Oatmeal Cookie	8.0	26	August 21, 2017
		8.1	27	December 5, 2017
Android Pie	Pistachio Ice Cream ^[26]	9	28	August 6, 2018
Android 10	Quince Tart ^[27]	10	29	September 3, 2019
Android 11	Red Velvet Cake ^[27]	11	30	September 8, 2020
Android 12	Snow Cone	12	31	October 4, 2021
Android 12L	Snow Cone v2	12.1 ^[a]	32	March 7, 2022
Android 13	Tiramisu	13	33	August 15, 2022
Android 14	Upside Down Cake ^[30]	14	34	October 4, 2023
Android 15	Vanilla Ice Cream ^[31]	15	35	September 3, 2024
Android 16	Baklava ^[32]	16	36	June 10, 2025
Android 17	Cinnamon Bun ^[4] ^[33]	17	37	June 16, 2026

Legend: Unsupported Supported Latest version Preview version

API Levels

<https://apilevels.com/>

Version	SDK / API level	Version code	Codename	Cumulative usage ¹	Year ⁴
Android 17	Level 37.1 Android 17 QPR1 BETA	CINNAMON_BUN_1	Cinnamon Bun ²	0%	2026
	Level 37 Android 17	CINNAMON_BUN			
Android 16	Level 36.1 Android 16 QPR2	BAKLAVA_1	Baklava ²	22.3%	2025
	Level 36 Android 16	BAKLAVA			
	▪ targetSdk must be 36+ for new apps and app updates as of August 31, 2026.				
Android 15	Level 35	VANILLA_ICE_CREAM	Vanilla Ice Cream ²	41.0%	2024
Android 14	Level 34	UPSIDE_DOWN_CAKE	Upside Down Cake ²	54.5%	2023
Android 13	Level 33	TIRAMISU	Tiramisu ²	68.9%	2022
Android 12	Level 32 Android 12L	S_V2	Snow Cone ²	78.8%	
	Level 31 Android 12	S			
Android 11	Level 30	R	Red Velvet Cake ²	86.9%	2020
Android 10	Level 29	Q	Quince Tart ²	91.1%	2019
Android 9	Level 28	P	Pie	93.5%	2018
Android 8	Level 27 Android 8.1	O_MR1	Oreo	94.8%	2017
	Level 26 Android 8.0	O		96.1%	
Android 7	Level 25 Android 7.1	N_MR1	Nougat	96.4%	2016
	Level 24 Android 7.0	N		96.6%	
Android 6	Level 23	M	Marshmallow	98.0%	2015
	▪ Jetpack/AndroidX libraries require a minSdk of 23 or higher since June 2025. ▪ Google Play Services v24.28+ requires API Level 23 or higher since July 2024 .				
Android 5	Level 22 Android 5.1	LOLLIPOP_MR1	Lollipop	98.2%	2014
	Level 21 Android 5.0	LOLLIPOP, L		99.8%	
	▪ Jetpack/AndroidX libraries require a minSdk of 21 or higher since April 2024. ▪ Jetpack Compose requires a minSdk of 21 or higher. ▪ Google Play services v23.30.99+ (August 2023) drops support for API levels below 21.				

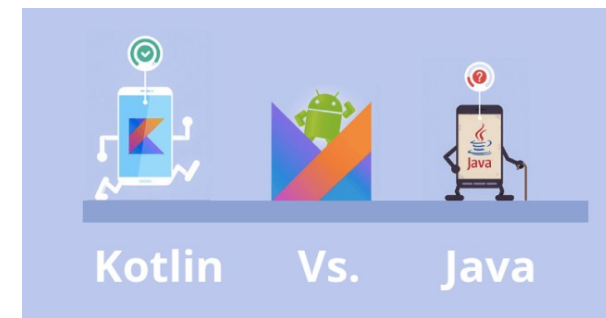
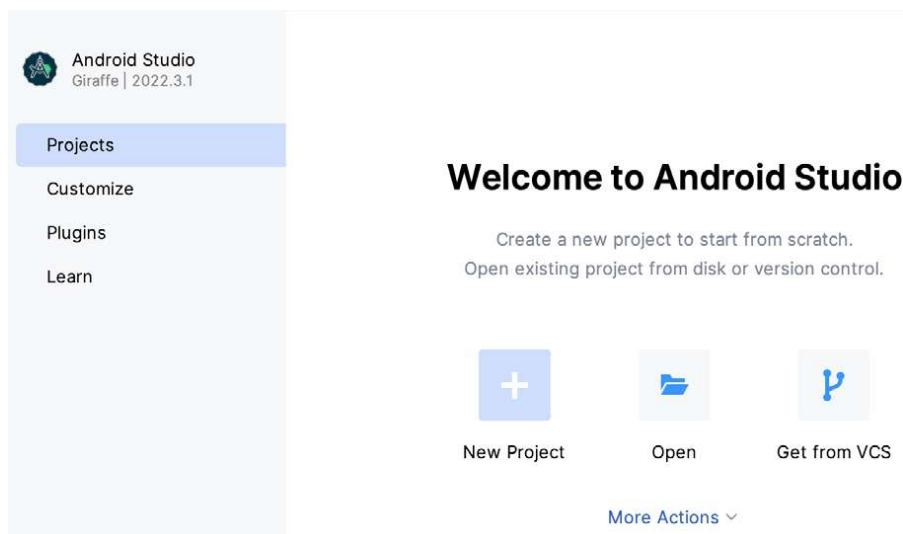
2. Setting up an Android Studio Development Environment

Inštalácia

System requirements

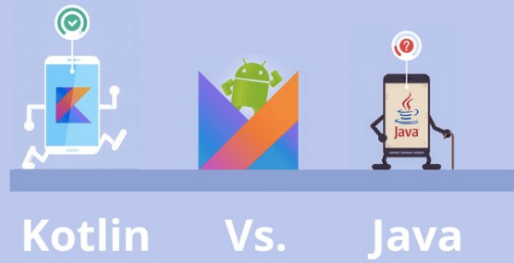
- Windows 8/10/11 64-bit
- macOS 10.14 or later running on Intel or Apple silicon
- Chrome OS device with Intel i5 or higher
- Linux systems with version 2.31 or later of the GNU C Library (glibc)
- **Minimum of 8GB of RAM**
- **Approximately 8GB of available disk space**
- 1280 x 800 minimum screen resolution

<https://developer.android.com/studio/index.html>



Android SDK Packages

Tools/SDK Manager tab SDK Platforms - API 37



Settings

Search

Appearance & Behavior

- Data Editor and Viewer
- Quick Lists
- Required Plugins
- Trusted Locations
- Path Variables
- Presentation Assistant

Keymap

- Editor
- Plugins
- Version Control
- Build, Execution, Deployment
- Languages & Frameworks
 - C/C++
 - Android SDK**
 - JVM Logging
 - Kotlin
 - Markdown
 - Schemas and DTDs
 - Template Data Languages
- Tools
- Backup and Sync
- Advanced Settings
- Other Settings
- Experimental
- Studio Labs

Languages & Frameworks > Android SDK [Revert changes](#)

Manager for the Android SDK and Tools used by the IDE

Android SDK Location: [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

Name	API Level	Revisi...	Status
☐ Sources for Android 37.1	37.1	1	Not install...
☐ 16 KB Page Size Google Play ARM 64 v8a System Image	37.1	8	Not install...
☐ 16 KB Page Size Google Play Intel x86_64 Atom System Image	37.1	8	Not install...
☐ 16 KB Page Size Google APIs ARM 64 v8a System Image	37.1	8	Not install...
☐ 16 KB Page Size Google APIs Intel x86_64 Atom System Image	37.1	8	Not install...
Android 17.0 ("CinnamonBun")			
☒ Android SDK Platform 37.0	37.0	2	Installed
☒ Sources for Android 37.0	37.0	2	Not install...
☐ Wear OS 7.0 ARM 64 v8a System Image (signed)	37.0	1	Not install...
☐ Wear OS 7.0 Intel x86_64 Atom System Image (signed)	37.0	1	Not install...
☐ Google APIs ARM 64 v8a System Image	37.0	6	Not install...
☐ Google APIs Intel x86_64 Atom System Image	37.0	6	Not install...
☒ Google Play ARM 64 v8a System Image	37.0	6	Not install...
☒ Google Play Intel x86_64 Atom System Image	37.0	6	Not install...
☐ 16 KB Page Size Google Play ARM 64 v8a System Image	37.0	6	Not install...
☐ 16 KB Page Size Google Play Intel x86_64 Atom System Image	37.0	6	Not install...
☐ 16 KB Page Size Google APIs ARM 64 v8a System Image	37.0	6	Not install...
☐ 16 KB Page Size Google APIs Intel x86_64 Atom System Image	37.0	6	Not install...

☒ Hide Obsolete Packages ☒ Show Package Details

OK Cancel Apply

Android SDK Packages

Tools/SDK Manager tab SDK Tools

The screenshot shows the 'Settings' dialog in Android Studio, with the 'Languages & Frameworks' section selected. The 'Android SDK' sub-section is active, and the 'SDK Tools' tab is chosen. The 'Android SDK Location' is set to 'C:\Users\borovan\AppData\Local\Android\Sdk'. A list of SDK tools is displayed with checkboxes for installation status. The 'Android Emulator' is highlighted. A 'Confirm Change' dialog is overlaid on the top right, listing components to be installed: Google Play ARM 64 v8a System Image API 37.0 (revision 6), Sources for Android 37.0 (revision 2), Android Emulator v.37.1.11, and Google Play Intel x86_64 Atom System Image API 37.0 (revision 6).

Settings

Appearance & Behavior

- Data Editor and Viewer
- Quick Lists
- Required Plugins
- Trusted Locations
- Path Variables
- Presentation Assistant

Keymap

Editor

Plugins

Version Control

Build, Execution, Deployment

Languages & Frameworks

- C/C++
- Android SDK**
- JVM Logging
- Kotlin
- Markdown
- Schemas and DTDs
- Template Data Languages

Tools

Backup and Sync

Advanced Settings

Other Settings

Experimental

Studio Labs

Languages & Frameworks > Android SDK [Revert changes](#)

Manager for the Android SDK and Tools used by the IDE

Android SDK Location:

SDK Platforms **SDK Tools** SDK Update Sites

Below are the available SDK developer tools. Once installed, the IDE will automatically check for updates. Check "show package details" to display available versions of an SDK Tool.

Name	Version	Status
☒ Android SDK Build-Tools		Update Available: 37.0.0
☐ NDK (Side by side)		Not Installed
☐ Android SDK Command-line Tools (latest)		Not Installed
☐ CMake		Not Installed
☐ Android Auto API Simulators	1	Not installed
☐ Android Auto Desktop Head Unit Emulator	2.0	Not installed
☒ Android Emulator	36.1.9	Update Available: 37.1.11
☒ Android Emulator hypervisor driver (installer)	2.2.0	Installed
☐ Android Project Templates	0.1.1	Not installed
☒ Android SDK Platform-Tools	36.0.0	Update Available: 37.0.1
☐ Android Support Repository	47.0.0	Not installed
☐ Google Play APK Expansion library	1	Not installed
☐ Google Play Instant Development SDK (Deprecated)	1.9.0	Not installed
☐ Google Play Licensing Library	1	Not installed
☒ Google Play services	49	Installed
☐ Google Repository	58	Not installed
☐ Google USB Driver	13	Not installed
☐ Google Web Driver	2	Not installed

☒ Hide Obsolete Packages ☐ Show Package Details

Confirm Change

The following components will be installed:

- Google Play ARM 64 v8a System Image API 37.0 (revision 6)
- Sources for Android 37.0 (revision 2)
- Android Emulator v.37.1.11
- Google Play Intel x86_64 Atom System Image API 37.0 (revision 6)

Cancel OK

OK Cancel Apply

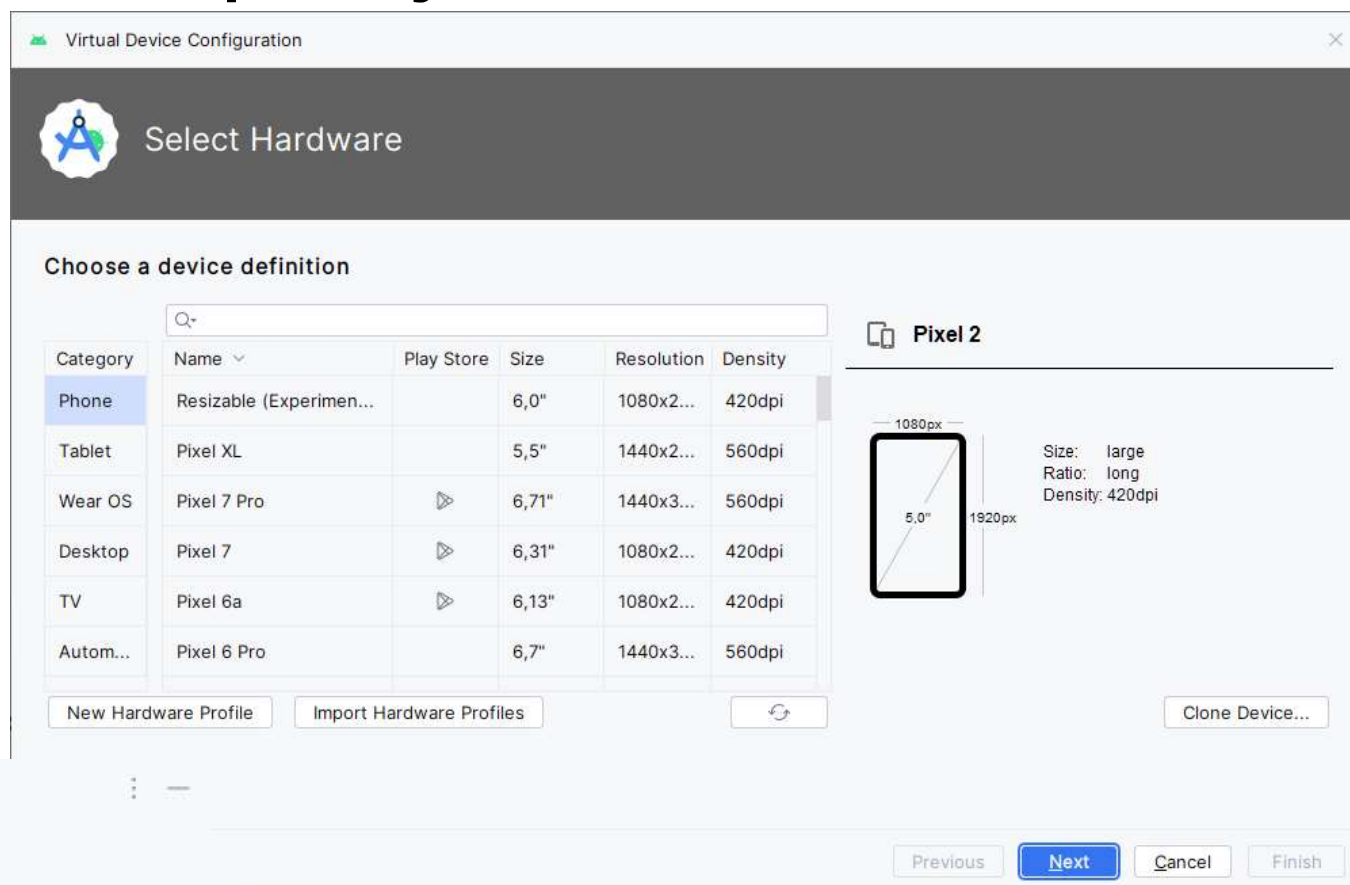
Android Virtual Device

Tools/AVD manager

Nakonfigurujte si AVD zodpovedajúci vášmu zariadeniu

alebo si vyberte zo zoznamu predvolených,

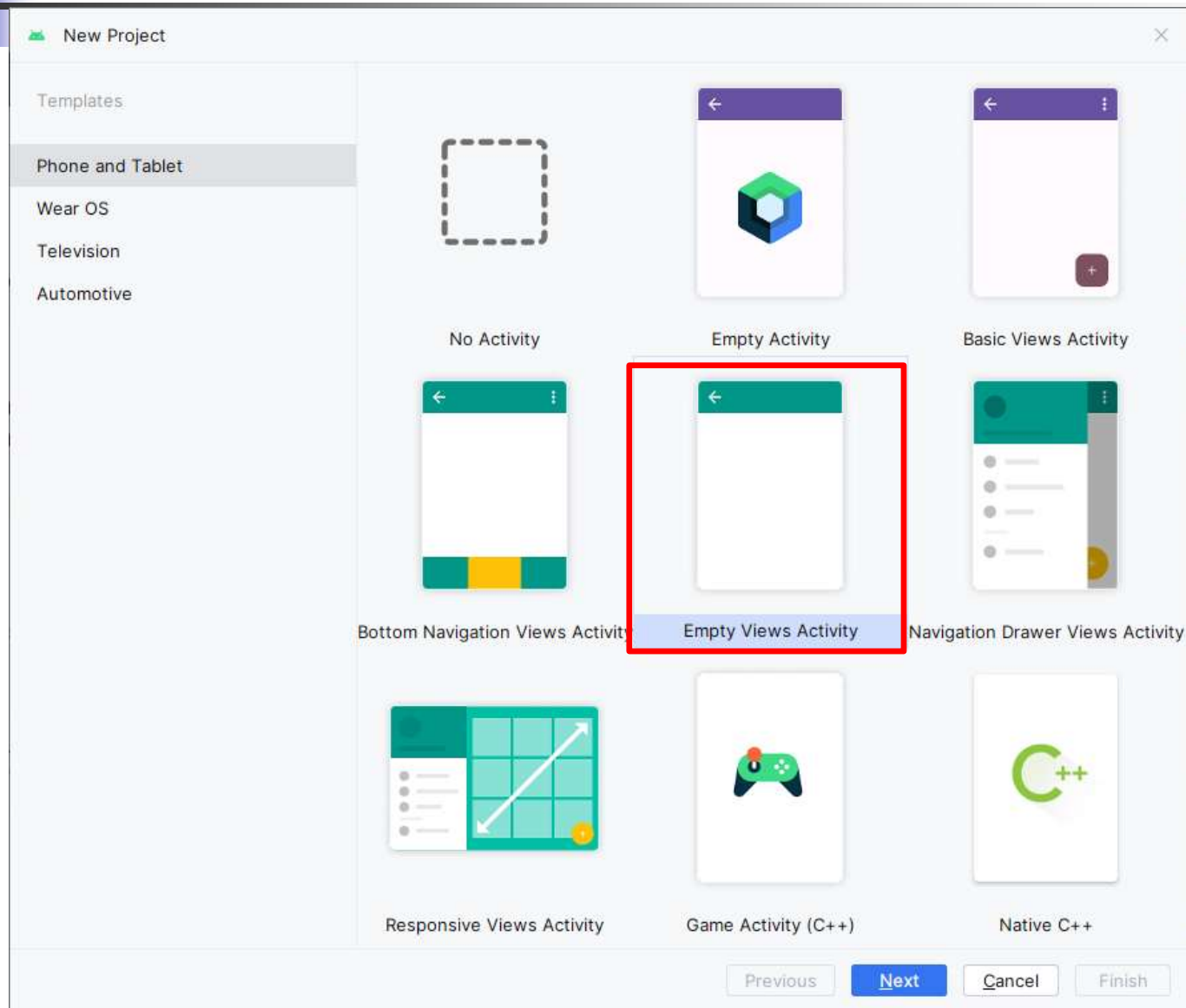
- Create Device
- modifikujte nastavenia podľa vášho zariadenia



3. Creating an Example Android App in Android Studio

Nový projekt

(File/New/New Android Project)



Nový projekt

(File/New/New Android Project)

★ New Project

Empty Views Activity

Creates a new empty activity

Name:

Package name:

Save location:

Language:

Minimum SDK:

i Your app will run on approximately **98,8%** of devices.
[Help me choose](#)

Build configuration language *?*

ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99.6%
4.2 Jelly Bean	17	98.1%
4.3 Jelly Bean	18	95.9%
4.4 KitKat	19	95.3%
5.0 Lollipop	21	85.0%
5.1 Lollipop	22	80.2%
6.0 Marshmallow	23	62.6%
7.0 Nougat	24	37.1%
7.1 Nougat	25	14.2%
8.0 Oreo	26	6.0%
8.1 Oreo	27	1.1%

Submitovanie riešení: Android SDK 17 (API 37),
(compileSdkVersion 37, buildToolsVersion "37.*"),
a min.požadované SDK (minSdkVersion 24)

API 34 ("UpsideDownCake"; Android 14.0)

i Your app will run on approximately **13,0%** of devices.
[Help me choose](#)

Nový projekt

(File/New/New Android Project)

```
plugins {  
    id("com.android.application")  
    id("org.jetbrains.kotlin.android")  
}
```

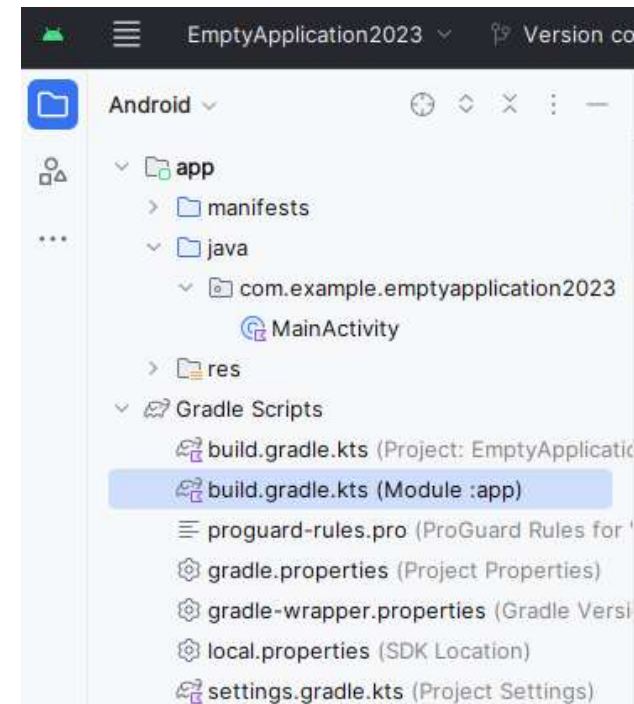
Submitovanie riešení: Android SDK 17 (API 37),
(compileSdkVersion 37, buildToolsVersion "37.*"),
a min.požadované SDK (minSdkVersion 24)

```
android {  
    namespace = "com.example.emptyapplication2024"  
    compileSdk = 37  
  
    defaultConfig {  
        applicationId = "com.example.emptyapplication2026"  
        minSdk = 24  
        targetSdk = 37  
        versionCode = 1  
        versionName = "1.0"  
    }  
}
```

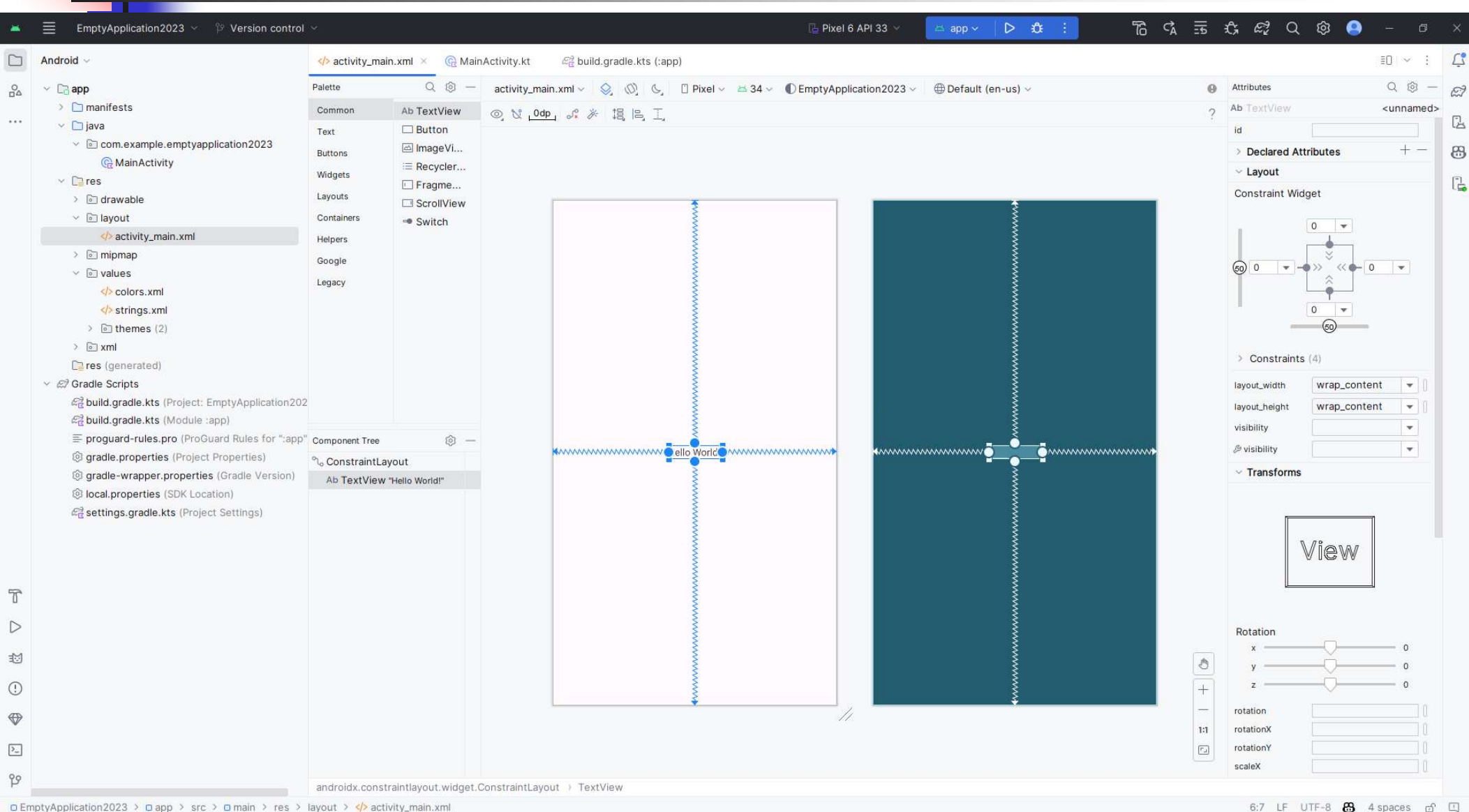
acts be like...



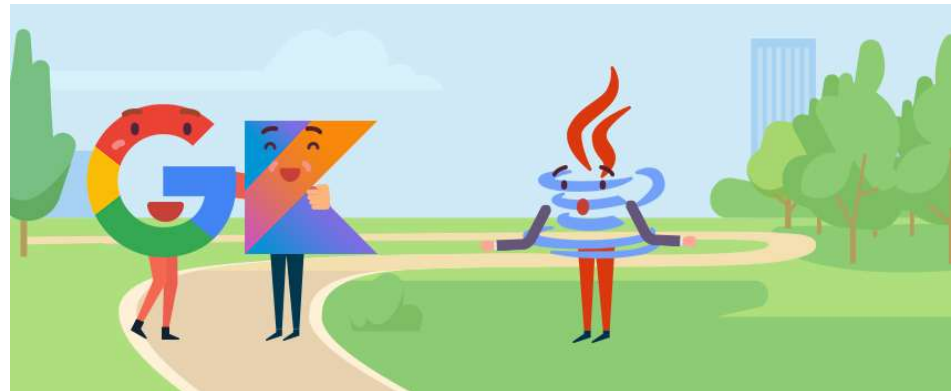
ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99.6%
4.2 Jelly Bean	17	98.1%
4.3 Jelly Bean	18	95.9%
4.4 KitKat	19	95.3%
5.0 Lollipop	21	85.0%
5.1 Lollipop	22	80.2%
6.0 Marshmallow	23	62.6%
7.0 Nougat	24	37.1%
7.1 Nougat	25	14.2%
8.0 Oreo	26	6.0%
8.1 Oreo	27	1.1%



Nový projekt (Empty views activity)



Pýtajte sa kým nedostanete



2023-08-28 09:57:15: Launching app on 'Pixel 6 API 33.'

```
$ adb shell am start -n "com.example.emptyapplication2023/com.example.emptyapplication2023.MainActivity" -a android.intent.action.MAIN -c android.intent.category.LAUNCHER
```

Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.example.emptyapplication2023/.MainActivity }

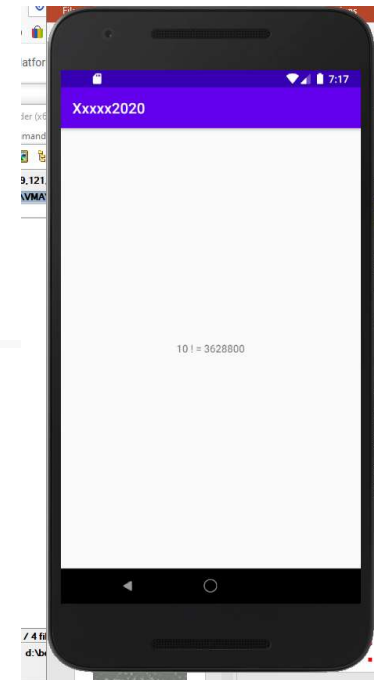
[Open logcat panel for emulator Pixel 6 API 33](#)

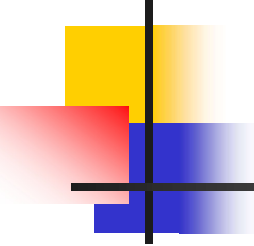
Connected to process 4535 on device 'Pixel_6_API_33 [emulator-5554]'.

Ako si skúšať Kotlin v AS

(kým sa nedozvieme viac)

```
class MainActivity : AppCompatActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContentView(R.layout.activity_main)  
        //println(fact(10))  
        for (i in 0..10) {  
            Log.d("TAG", "$i ! = ${fact(i)}")  
            // vypisuje do konzoly Logcat, použite filter s "TAG"  
            val tv = findViewById<TextView>(R.id.tv)  
            tv.text = "$i ! = ${fact(i)}"  
            // vypise do View komponentu, ktory je v Activite  
            Toast.makeText(this, "$i ! = ${fact(i)}",  
                Toast.LENGTH_SHORT).show()  
            // Toast alias Notifier (MITI)  
        }  
    }  
}  
  
fun fact(n : Int) : Int = if (n == 0) 1 else n * fact(n-1)  
}
```





Break point

(štruktúrou projektu pokračujeme na budúce)

- Switch to kotlin intro